Revision Notes

Class 6 Mathematics

Chapter 7 – Fractions

What have we discussed?

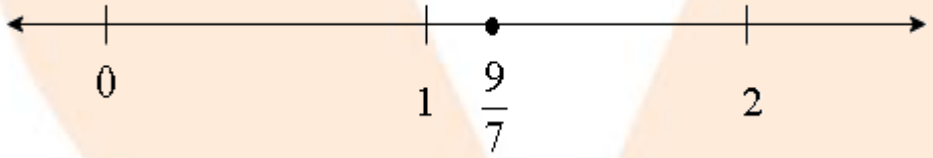
- A **fraction** is a number that represents a portion of something larger.
- A single object or a group of objects might make up the whole.
- It is necessary to guarantee that all parts are equal when stating a circumstance of counting components to write a fraction.
- As in example $\frac{8}{3}$, the numerator in this equation is 8 and the denominator is 3.

Fraction on the Number Line:

- A **number line** can be used to display fractions.
- Every fraction has a number line point connected with it.
- For example, how can we $\frac{9}{7}$ on a number line?

Solution:

The value of $\frac{9}{7}$ is 1.23 which means that $\frac{9}{7}$ lies between 1 and 2. We can show this on a number is as follows;



Like fractions are those that have the **same denominator**.

For example, $\frac{9}{30}, \frac{5}{30}, \frac{1}{30}$ are the **like fractions.**

Unlike fractions are fractions having **different denominators.**

For example, $\frac{3}{56}, \frac{2}{30}, \frac{8}{5}$ are the **unlike fractions.**

Proper Fraction:

- A proper fraction is a fraction that represents a portion of a whole.
- The **numerator of a proper fraction** is **less** than the **denominator**.

- For example, $\frac{3}{5}, \frac{7}{30}, \frac{6}{5}$ are the **proper fractions.**

Improper Fraction:

- **Improper fractions** are those in which the **numerator** is **higher** than the denominator.
- When an improper fraction is stated as a mixture of a whole and a part, it is referred to as a mixed fraction.
- There are several equivalent fractions for each right or improper fraction.
- A mixed fraction can be expressed as an incorrect fraction in the following way:

$$\frac{(\text{Whole} \times \text{Denominator}) + \text{Numerator}}{\text{Denominator}}$$

- For example, $4\frac{2}{5}, 3\frac{1}{9}, 12\frac{7}{8}$ are the **improper fractions.**

We can multiply or divide both the numerator and the denominator of a given fraction by the same number to discover an **equal fraction**.

For example, **Equivalent fraction** of $\frac{14}{56}$ is;

$$\frac{14}{56} = \frac{14 \div 14}{56 \div 14}$$

$$\frac{14}{56} = \frac{1}{4}$$

If the numerator and denominator of a fraction have no common factor except 1, it is said to be in the **simplest (or lowest) form**.

When adding or subtracting fractions, the LCM **of the denominators is selected as the common denominator.**

Comparing unlike fractions:

- **The fraction with the greater numerator is greater when two fractions have the same denominator.**

For example, if we compare $\frac{9}{3}$ and $\frac{4}{3}$ fractions, we can conclude that

$$\frac{9}{3} > \frac{4}{3}$$

That is

$$\Rightarrow \frac{9}{3} > \frac{4}{3}$$

$$\Rightarrow 3 > 1.33$$

- **When the numerator is the same in two fractions then, the fraction with the smaller denominator is greater of the two.**

For example, if we compare $\frac{7}{6}$ and $\frac{7}{3}$ fractions, we can conclude that $\frac{7}{6} < \frac{7}{3}$

That is

$\Rightarrow \frac{7}{6} < \frac{7}{3}$

$\Rightarrow 1.16 < 2.33$

How do we add or subtract mixed fractions?

- **Mixed fractions** can be expressed as a whole portion plus a proper fraction or as an improper fraction in its entirety.
- One method of adding (or subtracting) mixed fractions is **to do the operation separately for each of the whole parts.**
- The alternative method **is to write the mixed fractions as improper fractions and then add (or subtract) them directly.**
- **Example:**

  $4\frac{2}{3} + 3\frac{1}{4}$, We will solve the given example by applying both the methods.

  Solution:

  Method I:

  $$4\frac{2}{3} + 3\frac{1}{4} = 4 + \frac{2}{3} + 3 + \frac{1}{4}$$

  $$4\frac{2}{3} + 3\frac{1}{4} = 7 + \frac{2}{3} + \frac{1}{4} \qquad \text{-----}(1)$$

  Here,

  $$\frac{2}{3} + \frac{1}{4} = \frac{2\times4}{3\times4} + \frac{1\times3}{4\times3}$$

  $$= \frac{8}{12} + \frac{3}{12}$$

  $$= \frac{11}{12}$$

  $\frac{11}{12}$ can be written as;

$$\frac{11}{12} = \frac{12-1}{12}$$

$$= 1 - \frac{1}{12}$$

Therefore, equation (1) becomes,

$$7+\frac{2}{3}+\frac{1}{4} = 7 + 1-\frac{1}{12}$$

$$= 8 - \frac{1}{12}$$

$$= \frac{(12\times 8) - 1}{12}$$

Therefore,

$$7+\frac{2}{3}+\frac{1}{4} = \frac{96 - 1}{12}$$

$$7+\frac{2}{3}+\frac{1}{4} = \frac{95}{12}$$

Method II:

$$4\frac{2}{3}+3\frac{1}{4}$$

$4\frac{2}{3}$ can be written as

$$4\frac{2}{3} = \frac{(4\times 3)+2}{3}$$

$$4\frac{2}{3} = \frac{12+2}{3}$$

$$4\frac{2}{3} = \frac{14}{3}$$

$3\frac{1}{4}$ can be written as

$$3\frac{1}{4}=\frac{(3\times 4)+1}{4}$$

$$3\frac{1}{4}=\frac{12+1}{4}$$

$$3\frac{1}{4}=\frac{13}{4}$$

Therefore,

$$7+\frac{2}{3}+\frac{1}{4} = \frac{14}{3}+\frac{13}{4}$$

$$= \frac{(14\times 4)+(13\times 3)}{(3\times 4)}$$

$$= \frac{56+39}{12}$$

Since,
$(\text{LCM of 3 and 4} = 12)$

Therefore,

$$7+\frac{2}{3}+\frac{1}{4} = \frac{95}{12}$$